

RAGUL GOWTHAM H

Sustainability & Environmental Engineering | ISO 14001 EMS Lead Auditor | ESG, HSE & Renewable Energy | Research & Sustainability Professional

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PROFESSIONAL PROFILE

- **Energy & Environmental Engineering professional** with experience in sustainability, ESG/EHS, renewable energy, and environmental compliance.
- **ISO 14001:2015 EMS Lead Auditor** with practical experience in CSR consultancy, responsible sourcing, SEDEX/SMETA audits, and corrective-action management.
- **Research & Sustainability Professional** with interests and project experience in climate, hydrology, LCA, agrovoltatics, and controlled-environment agriculture.
- Experienced in connecting **international sustainability standards with practical field-level implementation and business operations.**

CORE COMPETENCIES

Sustainability & Environment	Environmental Sustainability · Environmental Management Systems · ISO 14001:2015 · Environmental Impact Assessment · Life Cycle Assessment · ESG · EHS · CSR · Responsible Sourcing
Social Compliance	Buyer Supplier Codes of Conduct · SEDEX / SMETA Audit Coordination · Corrective Action Plans · Supplier Engagement & Training · H&S Non-Conformity Identification · ETI Base Code · UN Global Compact · ILO Conventions
Renewable Energy	Solar PV Fundamentals · Hybrid Wind-Solar Systems · Agrivoltaics · Bioenergy · Feasibility & Project Framework Development
Research & Technical	Climate Variability & Rainfall Extremes · ENSO-IOD Teleconnections · Hydrology & Water Resources · FAO-56 Penman-Monteith Evapotranspiration Modelling · Solar Radiation Estimation · Experimental Design & Statistical Analysis · Controlled Environment Agriculture · DC Electrostimulation
Business Operations	Project Coordination & Reporting · DRHP / IPO Documentation Exposure · Statutory Filing Exposure (MGT-7, MGT-14, AOC)
Digital	Microsoft Office Suite · Power BI · MATLAB · PVSyst · HOMER · DSSAT · SolidWorks · Microsoft Copilot & AI Prompting

PROFESSIONAL EXPERIENCE

Technical & Business Operation Assistance Manager Progen Renewables Limited

20 May 2026 – Present

- **Corporate & Business Operations** — Gained working familiarity with IPO procedures and DRHP document preparation, and with the filing of essential statutory forms including MGT-7, MGT-14 and AOC-series returns.
- **ESG & EHS** — Applied EHS practices to the preparation of ESG reporting content in relation to SEBI disclosure requirements, supporting accuracy and traceability of reported information.
- **Technical Operations** — Monitored construction progress against quality-control standards and participated in routine maintenance under the supervision of experienced technicians; developed a working understanding of solar PV technology principles in an operational setting.
- **Reporting & Collaboration** — Prepared project status and performance-metric reports for internal stakeholders and contributed to a safe, productive working environment through cross-functional collaboration.

Learning Intern — Renewable Energy Project Development Progen Renewables Limited

June 2026 · One-week programme

- Developed project frameworks and feasibility models for hybrid wind and solar power generation systems.
- Conducted SWOT analyses to evaluate project viability and identify opportunities for process optimisation.
- Formulated strategic implementation and marketing plans supporting sustainable energy solutions.

Documentation Intern

Jan 2022 – Dec 2022

▲ Promoted from Intern

Jan 2021 – Jan 2022

Global Sustainability Solutions · CSR, Ethical Sourcing & Social Compliance

- Managed end-to-end CSR consultancy projects for global buyers, covering ethical sourcing, labour rights and responsible sourcing across the supplier base.
- Interpreted and implemented **Buyer Supplier Codes of Conduct (SCoCs)** across suppliers and vendors, translating buyer requirements into practical, site-level compliance actions.
- Planned and coordinated **SEDEX / SMETA audits** with suppliers, third-party auditors and internal teams, securing audit readiness and complete supporting documentation.
- Reviewed audit findings and developed **Corrective Action Plans (CAPs)** with suppliers, following through to closure to strengthen social-compliance performance; identified **Health & Safety non-conformities** during reviews and supported corrective measures.
- Delivered supplier training and capacity-building on ethical compliance and sustainability standards, applying the **ETI Base Code, UN Global Compact, ILO conventions** and applicable local labour laws.
- Liaised with buyers, suppliers and third-party auditors to safeguard transparency and integrity of compliance activities; reported key metrics and deliverables to internal stakeholders and clients while tracking evolving global CSR regulations and social audit standards.

FEATURED RESEARCH PROJECT

Potential of Electrostimulation in Enhancing Growth and Yield of Amaranthus spp. in Controlled Environment Agriculture

Project Team Member / Student Researcher · B.Tech. Energy and Environmental Engineering, Agricultural Engineering College and Research Institute, TNAU, Coimbatore

- Investigated **low-intensity direct-current (DC) electrostimulation** of the root zone as a growth-enhancement intervention for *Amaranthus* spp. under **Controlled Environment Agriculture (CEA)** conditions — at the intersection of electrical engineering, plant physiology and sustainable agriculture.
- Designed and instrumented the experiment (silver electrodes, regulated DC supply, milliammeter, multimeter) with treatment scheduling across crop-growth stages and matched control versus treated groups.
- **Characterised the applied stimulus by current density ($A\ m^{-2}$)** rather than raw voltage or current alone — a quantitative treatment definition making the electrical dose reproducible and comparable across replicates.
- Measured shoot and root length and biomass, Leaf Area Index and chlorophyll a, b and total chlorophyll, evaluating treatment effects statistically.

Reported outcomes: 22% increase in shoot biomass · 85% increase in root biomass · 46% increase in Leaf Area Index · increased chlorophyll a, chlorophyll b and total chlorophyll in treated plants.

RESEARCH PUBLICATIONS

Ragul Gowtham, H.*, Anto Rashwin, A., Ramachandran, J., & Rajkishore, S. K. (2026). **Evaluation of Hargreaves–Samani and Valiantzas Solar Radiation Models for FAO-56 Reference Evapotranspiration Estimation in Salem (Valapadi), Tamil Nadu, India.** *Madras Agricultural Journal*, 113(4–6), 198–210. DOI: 10.29321/MAJ.10.261330 · Accepted 1 June 2026.

**First and corresponding author — designed the study, analysed 25 years (2000–2025) of NASA POWER data and ran the model evaluations: Hargreaves–Samani reproduced measured ET_0 almost exactly (5.79 vs $5.78\ mm\ day^{-1}$); Valiantzas gave the stronger agreement ($R^2 = 0.982$).*

Ragul Gowtham, H., & Ramachandran, J. (2025). **Monsoonal Dynamics and Rainfall Extremes in Salem, Tamil Nadu, India: Insights from a Climate Variability Perspective.** *International Journal of Environment and Climate Change*, 15(7). Published 27 June 2025. DOI: 10.9734/ijecc/2025/v15i74911

Technical article — Ragul Gowtham, H., & Ramachandran, J. (2026). **ENSO–IOD Coupling and Monsoon Response in the Indian Ocean Basin During 2014–2025.** *Agriculture & Food: e-Newsletter*, 8(2), 288–290. Article ID 72788. ISSN 2581-8317.

Synthesis of ENSO–IOD phase evolution using Niño-3.4 (ONI), the Dipole Mode Index and satellite SSTs against monsoon rainfall departures.

PROFESSIONAL CERTIFICATIONS & TRAINING

Certified Environmental Management System (EMS) Lead Auditor Professional — ISO 14001:2015 Glocert International Certifications (UK) Limited · Exemplar Global approved · Training 12–16 May 2025	May 2025
Environmental Impact Assessment (EIA) — 94% consolidated score (70/75 examination) Agricultural Engineering College & Research Institute, TNAU · 9–10 August 2025	Aug 2025
Introduction to HSE Engineering · Risk Management and Industrial Safety · Environmental Protection and Sustainability Coursera · Authorized by Khalifa University · Completed 25–26 August 2026	Aug 2026
Workshop on Life Cycle Analysis on Agri-Food and Bioenergy Tamil Nadu Agricultural University	18 Sep 2025
UK–India Workshop on Agrophotovoltaic System Development for Net-Zero Farming Teesside University / TNAU collaboration	3 Mar 2025
Microsoft Copilot: The Art of Prompt Writing · LinkedIn Learning · AI Prompting & Productivity	2 Dec 2025
Android Mobile Application Development · SWAYAM · 4-credit course · 75% (proctored examination)	Dec 2024

AWARDS, CONFERENCES & LEADERSHIP

1st Prize — Eco Battle, ECOFEST’24 International Conference “Sustainable Synergy: Navigating the Energy and Environmental Nexus” · Tamil Nadu Agricultural University	23 Aug 2024
Organizing Committee Volunteer — ICASF’25 International Conference on Agrovoltatics and Sustainability in Farming · TNAU with Teesside University, UK · Also attended as conference participant	5–6 Aug 2025
Environmental & Community Engagement World Environment Day 2025 “Litter Lift Life” — Rotaract Club of TNAU with Coimbatore Wildlife Conservation Trust (5 Jun 2025) · Wildlife Patrolling — CWCT with TN Forest Department (14 Feb 2025) · Cyclothon, World Zero Emission Day — ECO CLUB, AEC&RI (24 Sep 2023)	2023 – 2025

EDUCATION & LANGUAGES

Bachelor of Technology (B.Tech.) — Energy and Environmental Engineering	Tamil	Mother tongue
Agricultural Engineering College and Research Institute	English	Read, write, spoken
TNAU, Coimbatore, India · Graduation: December 2026	Hindi	Read, spoken — learning
	Spanish	Read

Professional & research interests: EMS and ESG · HSE/EHS and risk management · LCA and EIA · solar PV and agrivoltatics · climate variability and monsoon dynamics · hydrology and evapotranspiration · responsible sourcing.